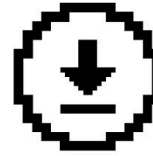


## 8 Improve the pattern

Now that you've changed the colour and size of the circle, you can try a few more things to improve the pattern. Let's give it a zany twist by changing the angle and position at which each circle is drawn. Make the changes highlighted in the code below, then run the program and see what happens.



Don't forget to save your work.

```
def draw_circle(size, angle, shift):
    turtle.pencolor(next(colors))
    turtle.circle(size)
    turtle.right(angle)
    turtle.forward(shift)
    draw_circle(size + 5, angle + 1, shift + 1)

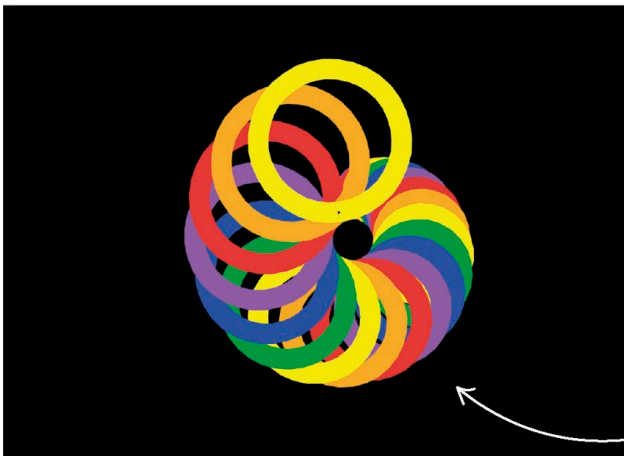
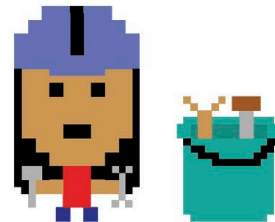
turtle.bgcolor('black')
turtle.speed('fast')
turtle.pensize(4)
draw_circle(30, 0, 1)
```

Annotations:

- Add these new parameters. (points to `angle, shift` in the function definition)
- The turtle turns clockwise. (points to `turtle.right(angle)`)
- The turtle moves forward. (points to `turtle.forward(shift)`)
- The angle and shift increase with every circle drawn. (points to `angle + 1, shift + 1` in the recursive call)
- Set the starting values of the new parameters. (points to `0, 1` in the initial call)

## Hacks and tweaks

Once everything is working smoothly, you can play around with the code and make the patterns even more fantastic.



### ◁ Chunky pen

Try increasing the pen size and see what it does to your pattern. You originally set it to 4 with the code below. What would 40 look like?

```
turtle.pensize(40)
```

The circles become chunkier when you increase the pen size.